General video to anime video translation

Presented as a part of IE 643 Course Project:

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Introduction

Problem Statement: Video to Anime Video Translation

Goal: Convert real-life videos into anime-style animation

Relevance: Growing interest in stylized video transformation, potential use in media production, anime industry



Background & Motivation

Motivation:

- Importance of Anime in Global Media
- High Production Costs

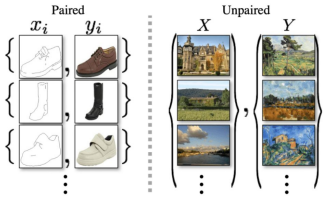
Benefits of Automation:

- Reduce Manual Work
- Increase Creativity
- Democratize Creation

Literature Review(1)

Pix2pix Model:

- **Type**: Supervised Generative Adversarial Network (GAN)
- **Purpose**: Image-to-image translation with paired data.
- **Application**: Transforms an input image to a target image (e.g., turning sketches into colored images).
- **Key Feature**: Requires paired datasets for training (input and target image pairs).



CycleGAN Model:

- **Type**: Unsupervised GAN
- **Purpose**: Image-to-image translation without paired data.
- **Application**: Converts images from one domain to another (e.g., turning photos into paintings or vice versa).
- **Key Feature**: Works with unpaired datasets by using a cycle consistency loss to ensure transformations are reversible.

Literature Review(2)

CartoonGAN Model:

- **Type**: Unsupervised Generative Adversarial Network (GAN)
- **Purpose**: Real-to-cartoon image translation without paired data.
- **Application**: Converts real-world images into cartoon/anime-style images.
- **Key Feature**: Does not require paired datasets for training.
- **Architecture**: Utilizes an encoder-decoder structure with **residual blocks** and edge-smoothness loss for stable and high-quality image generation.

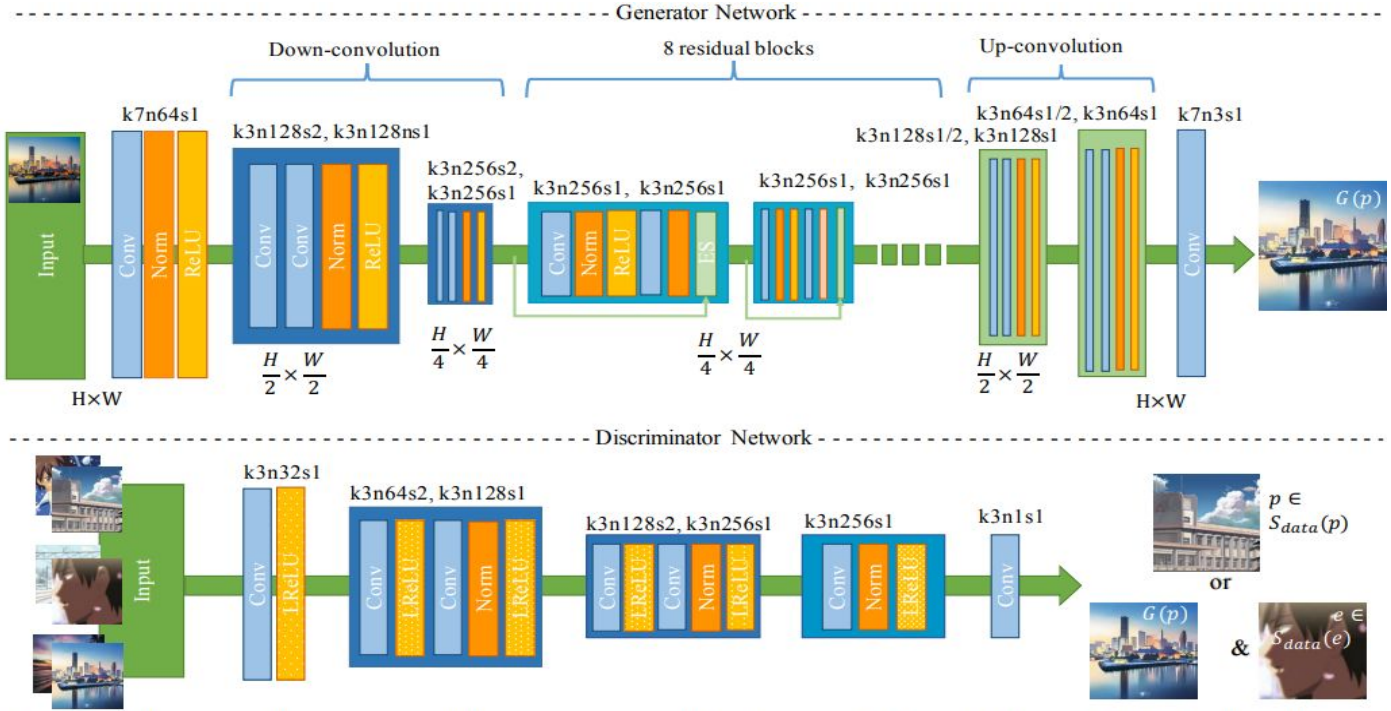


Figure 2. Architecture of the generator and discriminator networks in the proposed CartoonGAN, in which k is the kernel size, n i the number of feature maps and s is the stride in each convolutional layer, 'norm' indicates a normalization layer and 'ES' indicate elementwise sum

Problem Understanding

Core Challenges:

- Preserving **Temporal Consistency** across video frames
- **Anime Stylization**: Translating complex real-world visuals into simplified anime art
- **High Resolution Output**: Ensuring high-quality frame generation

What is AnimeGAN?

AnimeGAN Overview: A GAN designed to perform anime-style image generation

- **Architecture**: Generator-Discriminator model
- **Generator**: Translates real-world images into anime-style images
- **Discriminator**: Distinguishes between real anime and generated anime
- **Loss Functions**: Perceptual loss, Style loss, Adversarial loss



AnimeGAN Architecture

Key Components:

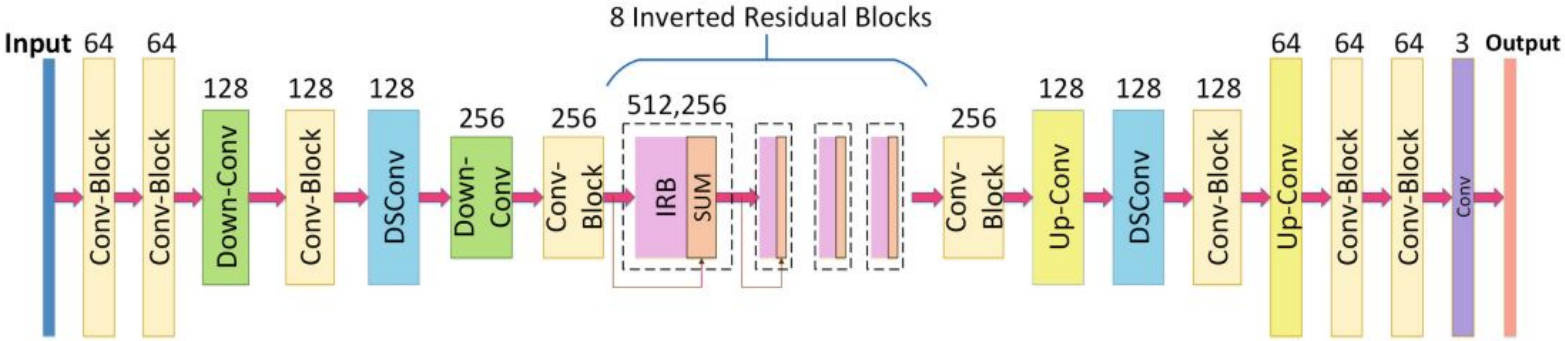
- **Generator**: Transforms input real-world images into anime-style outputs using a series of **convolutional layers** and **residual blocks** to preserve fine details and maintain structural integrity.
- **Discriminator**: Comprises a multi-layer convolutional network that assesses the realism of the generated images by distinguishing between genuine anime artworks and the outputs of the generator.
- **Loss Functions**: Adversarial Loss + Perceptual Loss + Style Loss for realistic anime effects

  Perceptual Loss: Utilizes high-level features from a pre-trained VGG network to ensure generated anime images retain the essence of the original input.

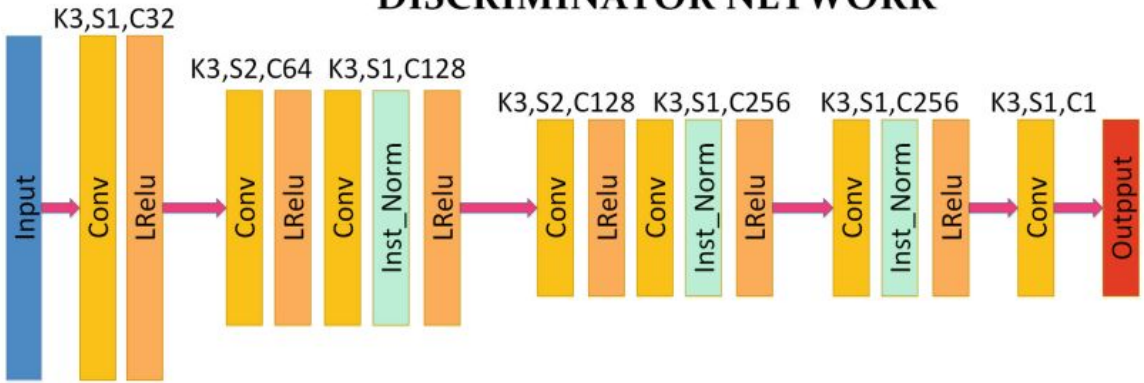
  Style Loss: Measures differences in style representations via Gram matrices to align the output with the distinctive aesthetics of anime.

  Adversarial Loss: Employs Wasserstein loss with gradient penalty for stable training, guiding the generator to create images that closely resemble real anime artwork.

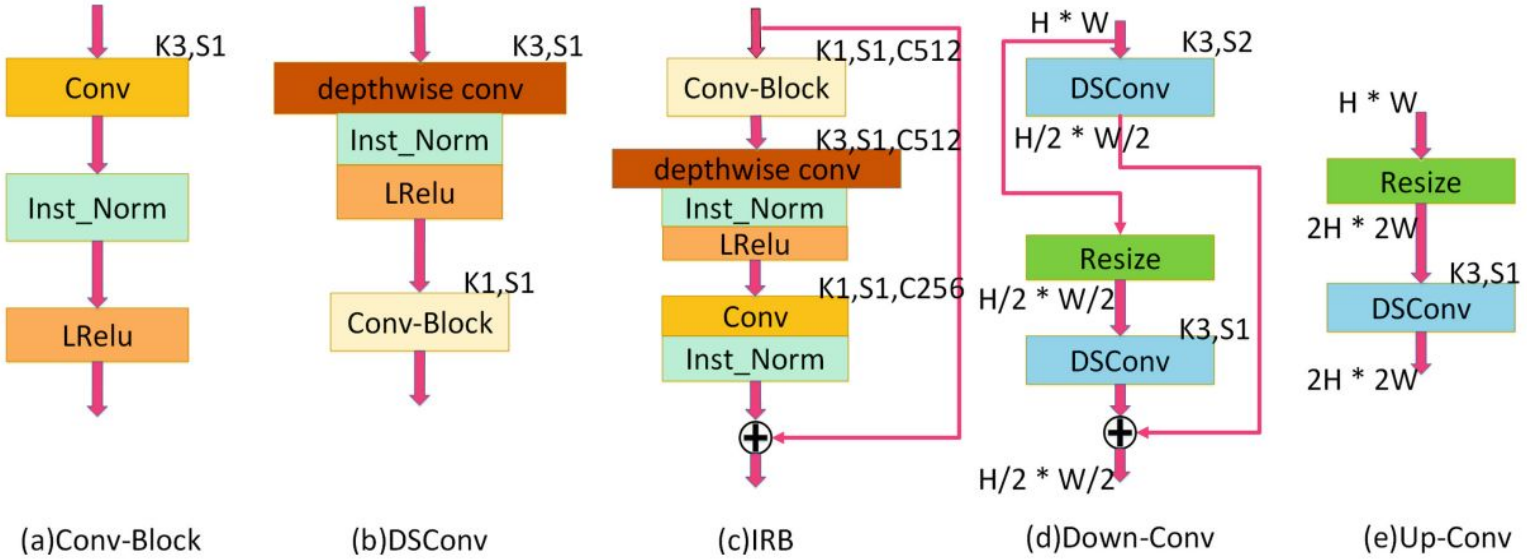
GENERATOR NETWORK



DISCRIMINATOR NETWORK



AnimeGAN



(a)Conv-Block (b)DSConv (c)IRB (d)Down-Conv (e)Up-Conv

Dataset Overview

CartoonGAN Dataset:

- **ImageNet**: Used for real-world image inputs to represent natural scenes.
- **Cartoon Datasets**: Consists of manually selected frames from popular anime films such as **Paprika**, **Howl's Moving Castle**, and **The Wind Rises**. These anime images serve as the style reference for training.

CycleGAN Dataset:

- **Photo-to-Van Gogh/Monet/Cezanne**: Real-world photos were paired with images from **Monet**, **Van Gogh**, **Cezanne**, and **Ukiyo-e** styles to test artistic style transfer.
- **Anime Datasets**: In some experiments, frames from anime movies like **Spirited Away** and **Princess Mononoke** were used to transfer realistic images into anime-style outputs.

Flowchart Workflow

Visual Overview of the process:

Input Video → Frame Extraction → Apply AnimeGAN → Frame Merging → Anime Video Output

- **Input Video**:
  - Begins with a video file that needs transformation into anime style.
- **Frame Extraction**:
  - The video is divided into individual frames for processing.
- **Apply AnimeGAN**:
  - Each frame is transformed using the AnimeGAN model to achieve an anime-style look.
- **Frame Merging**:
  - Transformed frames are reassembled in sequence to maintain the original motion.
- **Anime Video Output**:
  - The final result is an anime-style video that reflects the essence of the original input.

Dataset for fine tuning

1. **Real-World Video Dataset**:
   - **Source**: Utilizes high-quality clips from platforms like YouTube and open-source repositories.
   - **Content**: Features diverse real-world scenes and contexts to provide varied input for style transfer.
2. **Anime Video Dataset**:
   - **Source**: Curated collections of frames from renowned anime sources, such as Studio Ghibli films or other carefully selected anime clips.
   - **Content**: Captures distinct artistic styles, color palettes, and character designs representative of the anime genre.
3. **Relevance**:
   - **Training Purpose**: The combination of natural video data and anime frames enables AnimeGAN to learn effective mappings between realistic content and anime aesthetics.
   - **Enhancing Stylization**: This dataset supports the model in achieving high fidelity in video stylization by grounding its transformations in both realistic and artistic representations.

Initial Experiments

CycleGAN:

- **Goal**: Tested for real-to-anime image translation using unpaired image datasets.
- **Outcome**: Showed promising results but required complex training and lacked speed for real-time video processing.
- **Challenges**: Struggled with maintaining fine details in images and required additional post-processing to reduce artifacts.

CartoonGAN:

- **Goal**: Focused on converting real-world images into cartoon-style outputs.
- **Outcome**: Faster and more efficient than CycleGAN, with better performance in maintaining cartoon-like edges.
- **Challenges**: Limited by the quality of edge-smoothness in complex images and still not ideal for real-time video stylization.

Subtasks Breakdown

1. **Data Collection & Curation :**Collect video frames from real-world and anime datasets
2. **Preprocessing :** Frame extraction, resizing, normalizing
3. **Fine-tuning AnimeGAN :** Train on both anime frames and real video frames
4. **Model Testing :** Generate anime-style videos using the fine-tuned model
5. **Evaluation :** Evaluate style quality and temporal consistency

References

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Datasets

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2. https://www.kaggle.com/datasets/diraizel/anime-images-dataset